

University of Waterloo

Department of Mechanical and Mechatronics Engineering



Design of a Safety Vent Valve

Prepared for
Waterloo Rocketry Team

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May 18, 2020

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Prof. Andrew Kennings
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Dear Professor Kennings,

This report entitled “Safety Vent valve”, was prepared as my 2B work term report. This is my 1st work report submission. The purpose of the report is to document the design process of a safety vent valve to be included as part of a rocket engine.

The Waterloo Rocketry Team is a student design team at the University of Waterloo. The goal of the team is to design and manufacture a rocket to compete at the annual Spaceport America Cup hosted by Spaceport America and the Experimental Sounding Rocket Association. The team also aims to provide the students with hands on engineering experience and skills which can be valuable beyond the work done on the team.

This report was written entirely by me and has not received any previous academic credit at this or any other institution. I thank Nicholas Christopher, Zachariah Mears, Alex Mihaila, and other Waterloo Rocketry Team members for providing their time and experience to help in the development of the vent valve. I received no other assistance.

Best Regards,

A handwritten signature in black ink, appearing to read 'M Gencher', with a stylized, cursive script.

Matthew Gencher
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Summary

This report documents the design of a safety vent valve for the Waterloo Rocketry Team to be included in the next rocket the team enters in the Spaceport America Cup. The vent valve has the following functions: remains closed so the run tank remains a sealed pressure vessel, is controlled through radio to open if an issue occurs where the run tank needs to be vented of any nitrous oxide within it. The existent vent designs suffer from many issues which are addressed by the new design.

Based on the existing vent design, a new vent valve assembly was designed with fixing those deficiencies in mind. A solenoid valve was chosen to replace the previously used linear actuator. This change allowed the valve to default to a safe position of power was lost to the valve. Additionally, the circuit board required to control the valve was simplified by using a solenoid valve. The specific valve chosen, a Peter Paul Series 20 Model H21, was chosen based on its cost, weight, required current and housing material. These criteria were compared to other solenoid valves that met the required specifications needed. The circuit board was designed using the existing board which controls the injector valve.

It is concluded that the redesigned vent valve sufficiently addresses the issues with existing designs and will sufficiently meet the safety requirements set by the Experimental Sounding Rocket Association.

1 Introduction

The Waterloo Rocketry Team is a student design team at the University of Waterloo in Waterloo, Ontario. The goal of the team is to design and manufacture a rocket to compete at the annual Spaceport America Cup (SAC) hosted by Spaceport America and the Experimental Sounding Rocket Association (ESRA).

For the past years, the team has developed a hybrid fuel rocket engine [1]. This type of engine using a solid-state fuel and a liquid oxidizer. The solid-state fuel used by the team is Hydroxyl-Terminated Polybutadiene (HTPB) and the oxidizer used is the nitrous oxide. The engine consists of two main components, the combustion chamber which holds the solid-state fuel and the run tank which is filled with the oxidizer. The component relevant to the report is the run tank assembly and more specifically is the safety vent component of the run tank.

As stated in the SAC rules, all hybrid engines must have a way to vent the oxidizer in case of a need to abort launch after filling the run tank [2]. In past years, to comply with this rule, the team included a constant vent hole which was used for the rocket Shark of the Sky (SOTS). A valve actuated by a linear actuator, shown in figure 1, was used for Unexploded Ordinance (UXO). Although these solutions have worked in the past there are multiple issues with them, outlined in Section 2.1 which a new design would fix.

The objective of this report is to document the selection of a commercial of the shelf (COTS) valve and the electrical design to control the valve for the Waterloo Rocketry Team's next submission to the Spaceport America Cup. As the current COVID-19 pandemic has delayed assembly of the vent valve, validation testing is not within the scope of this report.

2 Design Specifications

This section will outline the use of the vent valve, past designs, design guidelines and the criteria and constraints that guided the development of the new vent valve design.

2.1 Existing Vent Valves

In past years, the Waterloo Rocketry Team had two designs for the vent valve. In 2018, UXO used a linear actuator to actuate a valve which allowed the run tank to vent when opened, shown in Figure 1.

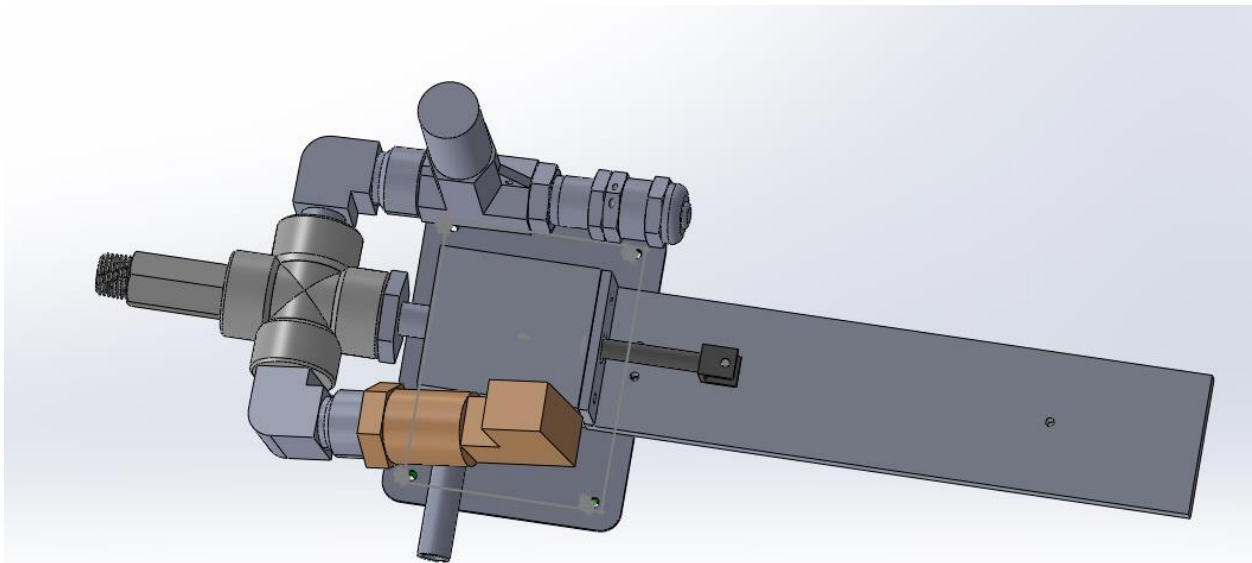


Figure 1: Linear Actuator Vent Valve

Although this assembly flew in UXO, there existed multiple issues with it that had to be addressed. The most prominent issue was a safety issue that had to be addressed in the next iteration of the vent valve. In an email with the manufacturer of the valve, a Ian Greig, a Sales Representative for Actuonix Motion Devices said, “If a P16-P (valve) loses power, it will hold position unless the force on it exceeds it's back drive rating” [3]. This means that if radio connection to the rocket was lost once it was filled there would be no way to safely open the valve and vent the run tank.

The board which was used to control the vent valve in 2018 had no inherent flaws in of itself. However, due to the use of a linear actuator to control the opening and closing of the valve, the board had a lot of components, and by virtue of that was quite large. Additionally, it was more prone to failure as there were more components which could be the cause of a problem. Figure 2 shows the schematic of the section of the board which controlled the opening and closing of the valve. The complexity of this design will be contrasted to the new design in Section 3.2.

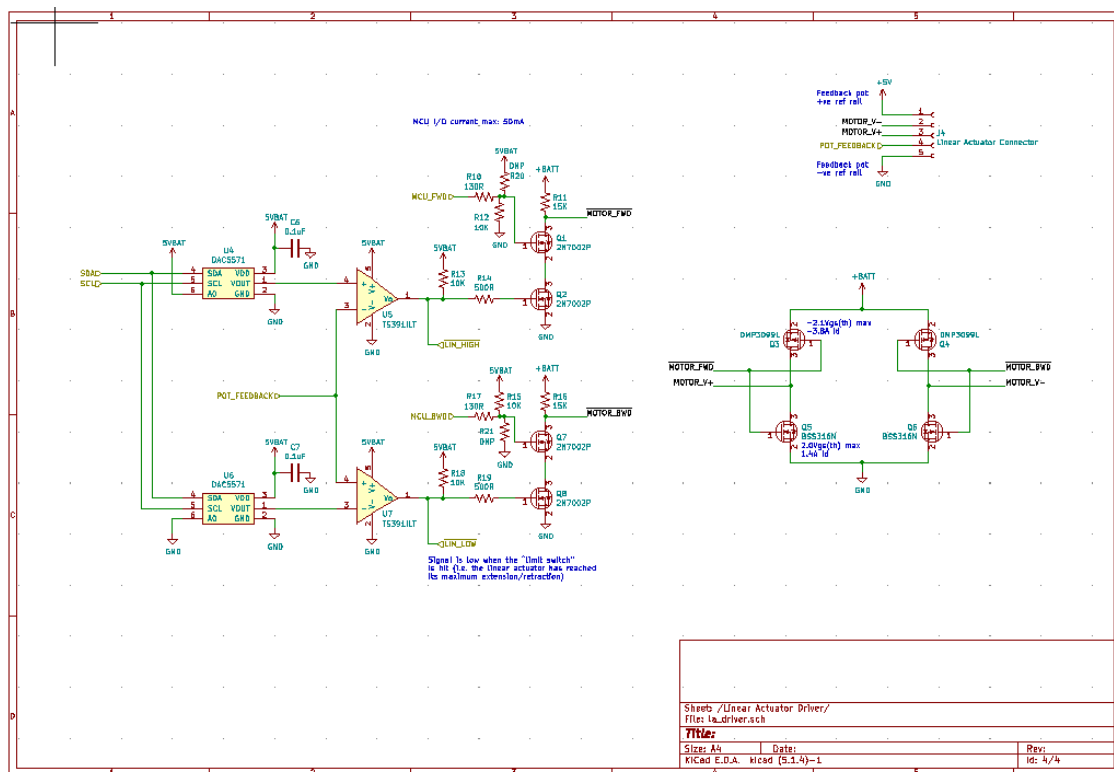


Figure 2: Linear Actuator Control Schematic

In 2019, SOTS moved away from using an electrically actuated valve and had a constant 1mm hole in the run tank bulkhead. Although this design allows the tank to vent without a radio connection to the rocket it comes with its own issues that needed to be improved upon in the next iteration of the vent valve. Since the run tank constantly vents once it has been filled, the longer it takes between filling and launch, the more nitrous oxide is being lost, which translates to a lower apogee. Additionally, since the run tank was not perfectly sealed, the maximum pressure in the tank is limited, which is a problem when the pressure of the nitrous oxide contributes to initial impulse of the rocket.

2.2 Criteria and Constraints

Since the report outlines two components of the vent valve assembly the criteria and constraints for each component will be listed separately. The constraints for the selection of the COTS valve are listed as follows with explanations as appropriate:

1. The valve must fail open so the tank will start to vent if power is lost to the rocket while it is on the launch pad.
2. The valve must be rated to at least 1000psi.
3. The valve must be rated for use with oxygen, so the nitrous oxide does not degrade its quality.
4. The valve must work in a temperature range of -20°C to 35°C so it can be used in testing in the winter as well in the desert in the summer during competition.
5. The valve must be able to operate off a 12V DC power source so it can be powered by a single 12V lithium-ion battery.

The constraints for the selection of the COTS valve are listed as follows with explanations as appropriate:

1. The valve should be a low cost.
2. Weight should be minimized to keep the overall rocket weight as low as possible.
3. The valve should be able to integrate with the existing plumbing. (Meaning the ports should be 1/8" NPT).
4. The valve should draw a low operating current to increase battery life.
5. The orifice size in the valve should be as large as possible to allow the run tank to vent as quickly as possible.
6. The material of the valve's housing should allow it to have as long a lifespan of possible.

The constraints for the electrical design are listed as follows:

1. The board must be integrated with the Controller Area Network (CAN) bus protocol that is used in the rest of the rocket.

2. The board must allow full control over the selected valve using a micro-controller.
3. The board must contain a fuse in case the battery starts to discharge too much current.
4. The board must be able to integrate with the selected valve.

The criteria for the electrical design are listed as follows:

1. The board should be as small as possible to minimize additional weight.
2. The board should be able to be mounted on the same hardware as the other electronics in that section of the rocket.
3. The cost of the board and the components should be minimized.
4. The board should contain ways to debug the code such as LEDs and limit switches.

3 Discussion

Figure 3 shows a drawing of the selected COTS solenoid valve. Figure 4 shows a three-dimensional view of the printed circuit board (PCB) for the board that controls the valve and the schematic can be found broken down into its sheets in figures 5, 6, 7 and 8 in section 3.2. The aspects of valve selection and of the circuit board design are outlined within this section.

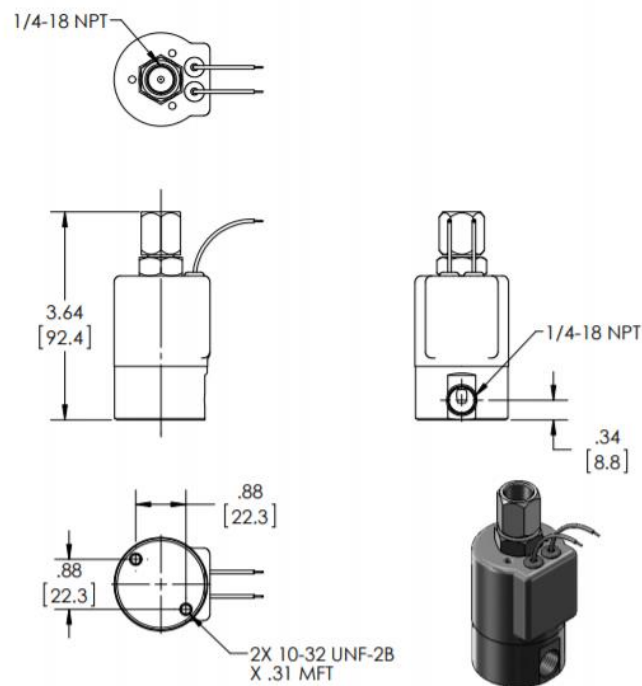


Figure 3: Selected Solenoid Valve Drawing [5]

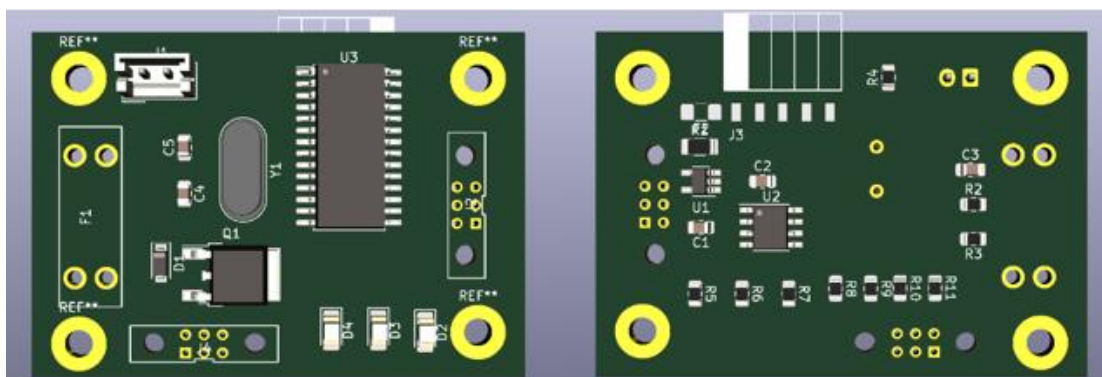


Figure 4: Front and Back Views of the Designed PCB

3.1 Valve Selection

Due to the constraints the valve had to meet; it was decided that a solenoid valve would be used for the application. This decision was made due to the versatility of what media solenoid valves can handle, as well as that a solenoid valve can fail open [4], an essential feature to meet the safety requirements. Table 1 outlines, multiple valves that were chosen has possible options, as well as how they fit the outlined criteria in section 2.2. Each of the below valves met the required constraints.

Table 1: Solenoid Valve Options

Manufacturer	Model Name	Holding Current	Weight (lbs)	Orifice Size	Housing Material	Cost
Clarke Cooper	EH40-04-DO12-NOOXXP [6]	1.64A	3.1	0.25"	Stainless Steel	\$935
Peter Paul	Series 20 Model H21 [5]	0.8A	1.19	0.024"	Stainless Steel	\$350
ASCO	EE8262R161 [7]	0.34A	1.41	0.047"	Brass	\$362

Each of the above valves have their advantages and disadvantages. Although a decision matrix was not used, how each valve met the given criteria was used heavily in which valve was selected. Although the Clarke Cooper valve had the largest orifice size, meaning it would allow the run tank to vent in the least amount of time the other, the other factors outweighed this advantage. First, it weighed the most of the three selected options. Due to the nature the rocketry project, weight minimization is extremely important in achieving the maximum apogee possible. Additionally, the high holding current of the valve would have required the addition of a coil saving circuit. A coil saving circuit is a circuit that alternates the power to the valve between on and off which allows it stay closed while using only half the power that would be used normally [8]. The addition of a coil saver circuit would have made the electrical design for the circuit board more complex, as well as increased the size of the board as more components would have been required. The Peter Paul valve and the ASCO valve are very similar in the listed criteria. Although the ASCO valve has a lower holding current usage, with the 2200 mAh lithium-ion batteries used, a holding current of 0.8A allows the batteries to power the valve for 2.75 hours. Due to past experiences at competition, this battery life was long enough as the

rocket never stood on the launch pad after being filled for the long before launch or the need to abort. Due to the larger orifice size of the ASCO valve, it would allow the run tank to vent slightly faster, although this was not a major criterion considered. The big difference between these two valves is the material the valve is made off. The Peter Paul valve is made of stainless steel where the ASCO valve is made from brass. Due to this difference in material, the Peter Paul valve was selected as stainless steel would provide the valve with a longer life span. With the valve selected the next step in the design process was to design the electric circuit board that would control the valve.

3.2 Electrical Design

The electrical design for the vent valve was based around the circuit board for the injector valve, which is used in a different section of the rocket. The schematics for the injector valve can be found in Appendix A. The schematics for the vent valve are shown in figures 5, 6, 7, and 8. Figure 5 shows the connection between sheets.

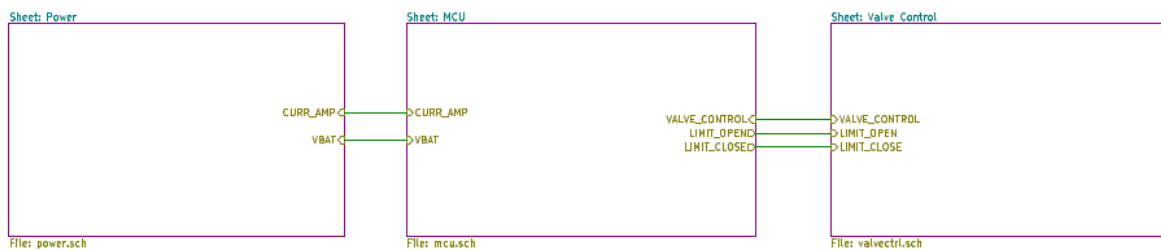


Figure 5: Connection Between Schematic Sheets

Figure 6 shows the power sheet of the vent valve schematic. Since both the vent valve and injector valve are powered by a single 12V lithium-ion battery, the power sections of the schematics are near identical, the only difference being the size of the fuse used. As a solenoid valve pulls much less current than the valve used for the injector, a 10A fuse was substituted for the 30A fuse used in the injector valve.

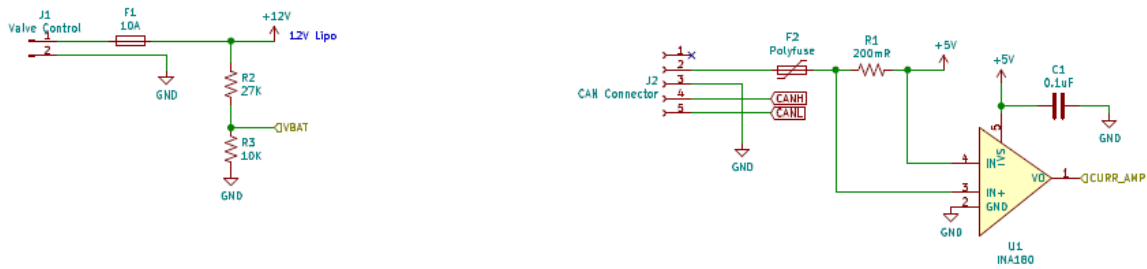


Figure 6: Power Sheet in Schematic

Figure 7 shows the sheet of the schematic which contains the microcontroller. A PIC18F23K83 was the microcontroller used for the board, as it is the same microcontroller used for other electrical sections within the team. Like the power section of the board, the vent valve is very similar to the injector valve within the section of the schematic. The difference between this section of the board and that of the injector valve is how the valve is connected to the board. Where the injector valve has six connections that need to be made to the board, the solenoid valve chosen only has two connections, power, and ground. Additionally in this section of the board is the oscillator which the microcontroller uses as a clock, a programming header to program the microcontroller, and a CAN header so the board can receive commands from the CAN protocol used in the rest of the rocket. There are also three LED's to be used in debugging once the code for the board is written and being tested.

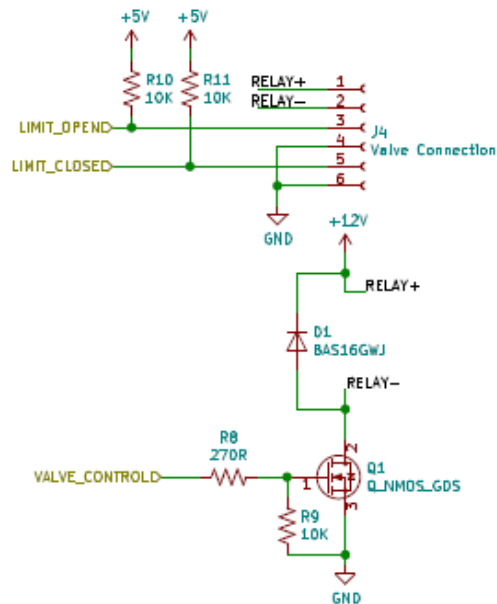


Figure 8: Valve Control Sheet in Schematic

After the schematic was planned and had gone through board review from multiple team members the components were plotted to a PCB, shown in figure 4. Once the board was plotted it went through a board review like the schematic review. That process was iterated through until the connections on the board were as simplified as possible and the size of the board was as small as possible. Then the board was printed, and the PCB was populated with all the required components, shown in figure 9.

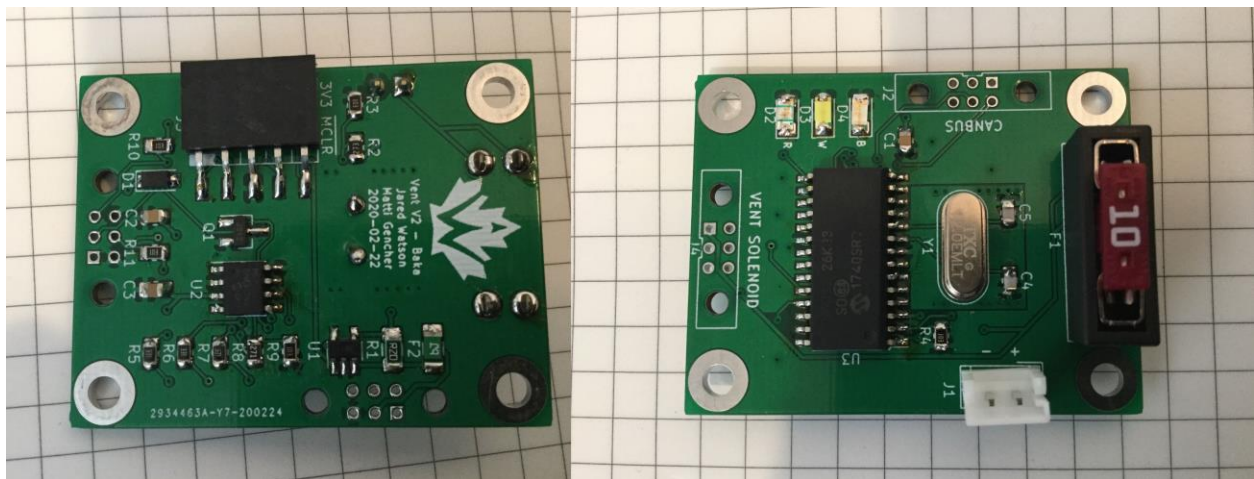


Figure 9: Assembled Circuit Board

4 Conclusions

The redesigned vent valve will be sufficient to meet the rules and guidelines supplied by ESRA to compete in the Spaceport America Cup. Additionally, it meets the other criteria required, such as low weight, low cost, and will integrate into the existing systems within the rocket.

The selected solenoid valve is compatible with high pressure nitrous oxide, which was a requirement for the project. Additionally, it adds minimal extra weight to the rocket and will last multiple years due to the material used in construction.

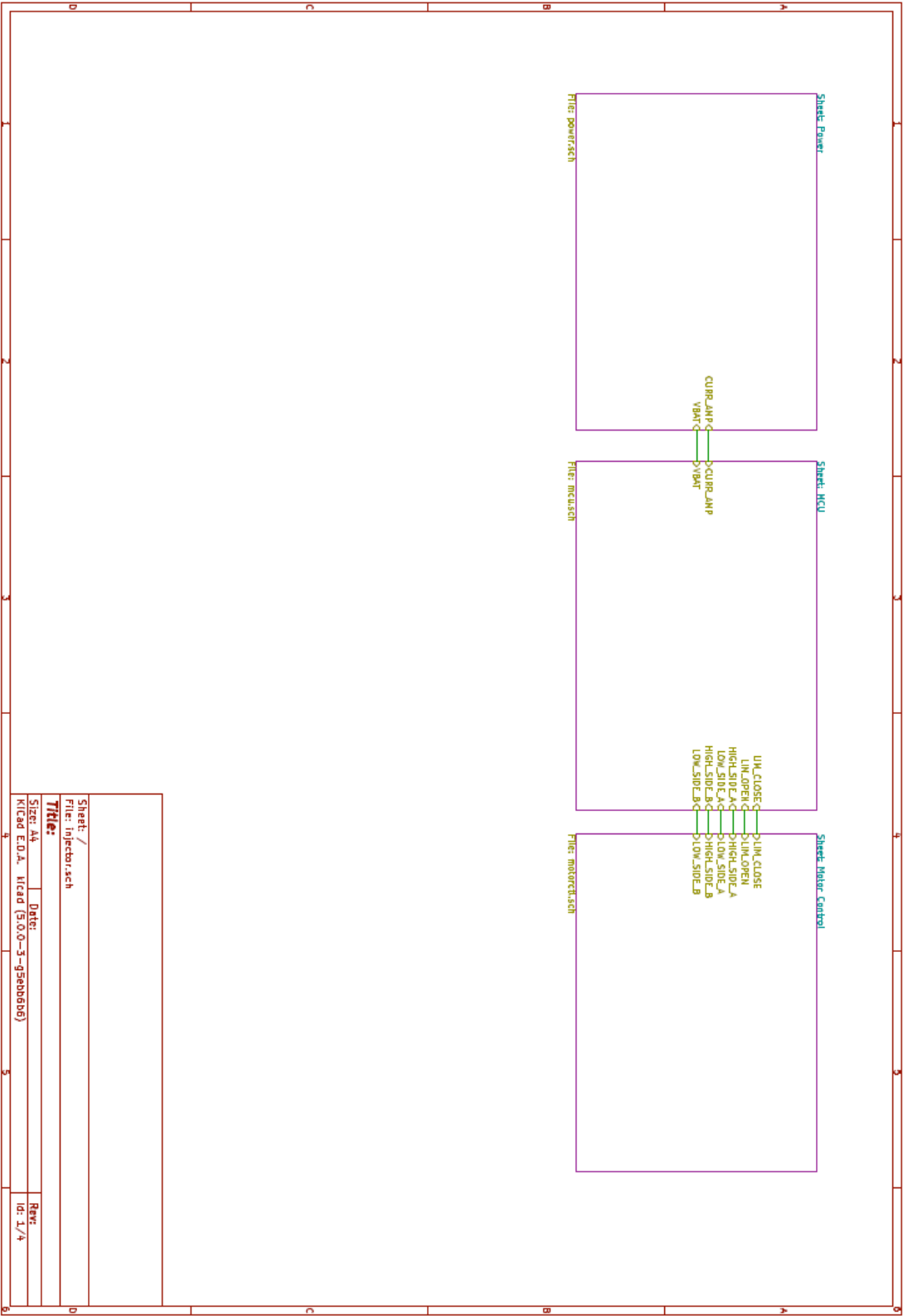
The modified injector board which will be used as the vent board is much simpler than the old board and will result in less risk of error or malfunction. The addition of LEDs and limit switches will allow easier debugging for any code written for the board.

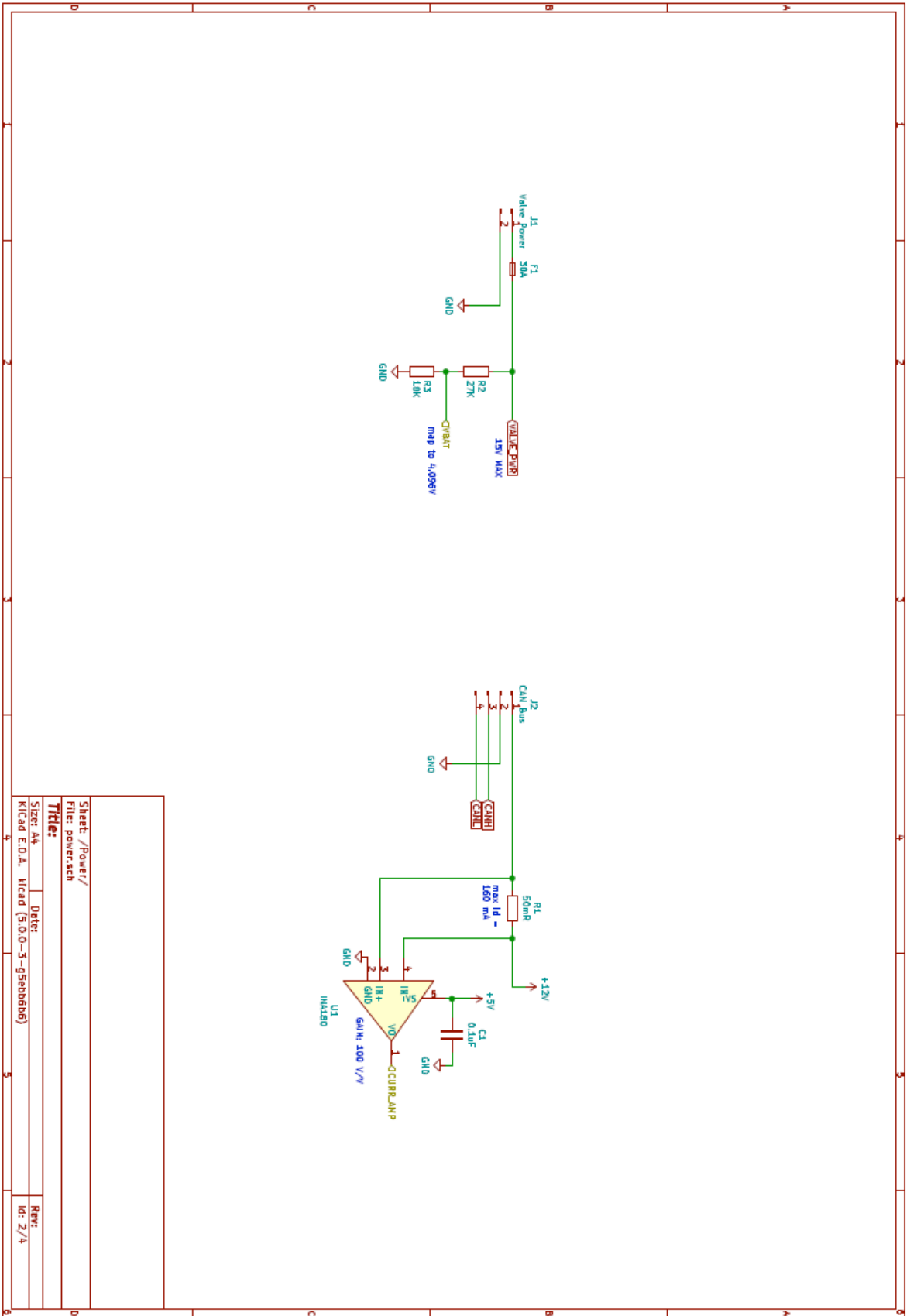
Overall, the redesigned vent valve is a suitable replacement for the systems that were used in the past. The redesign addresses and improves upon the problems past systems had.

References

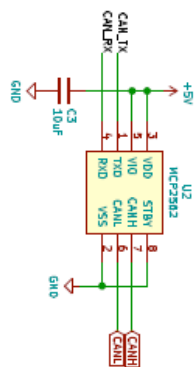
- [1] (2019) Rockets. Waterloo Rocketry Team. [Online]. Available: <http://waterloorocketry.com/rockets.html>
- [2] ESRA, “SA Cup IREC Rules & Requirements.” 13-May-2018.
- [3] “Re: Linear Actuator,” *Re: Linear Actuator*, 17-Sep-2019.
- [4] S. Peters, “What The Heck Is A Solenoid Valve?,” *Crane's Fluid Connection Blog*, 03-Sep-2015. [Online]. Available: <https://blog.craneengineering.net/what-the-heck-is-a-solenoid-valve>. [Accessed: 10-May-2020].
- [5] Peter Paul, “Series 20 Model H21,” *Series 20 Model H21 | Peter Paul*. [Online]. Available: <https://peterpaul.com/valves/2-way-normally-opened/series-20-model-h21>. [Accessed: 10-May-2020].
- [6] Clark Cooper, “The EH40 Series,” *EH40 Series Valves | High Pressure Solenoid Valve | Clark Cooper*. [Online]. Available: <https://clarkcooper.com/hpv/EH40.html>. [Accessed: 10-May-2020].
- [7] ASCO, “Electronically Enhanced Solenoid Valves,” *ASCO*. [Online]. Available: <https://www.asco.com/ASCO Asset Library/asco-next-gen-valves-catalog.pdf>. [Accessed: 10-May-2020].
- [8] Axiomatic Technologies Corporation, “Hot Shot Coil Saver.” [Online]. Available: <https://www.axiomatic.com/TD2100AX.pdf>. [Accessed: 10-May-2020].\

Appendix A: Injector Valve Schematic





Sheet: /Power/	
File: power.sch	
Title:	
Sheet: A4	Date:
Kicad E.O.A. Kicad (5.0.0-3-g5eb0b5b)	
Id: Z/4	Rev:



Sheet: /MCU/	
File: mcu.sch	
Title:	
Size: A4	Date:
Kicad E.D.A. v14.0 (3.0.0-3-g5eb0ab5)	
Rev:	Rev:
Id: 5/4	

